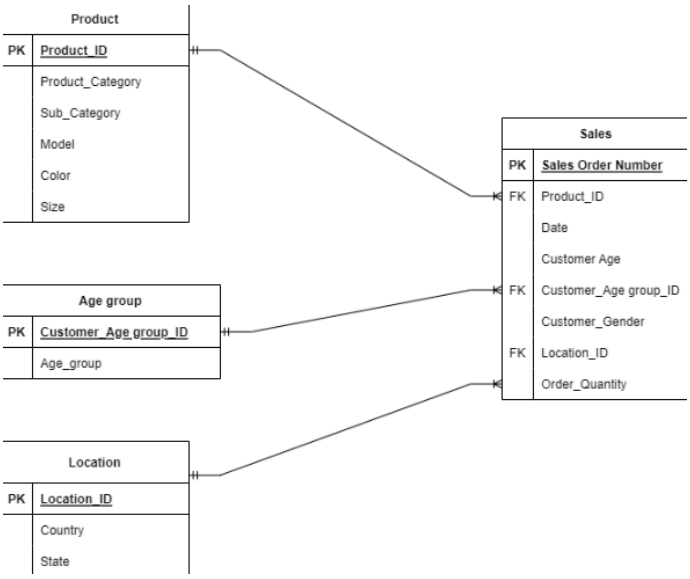
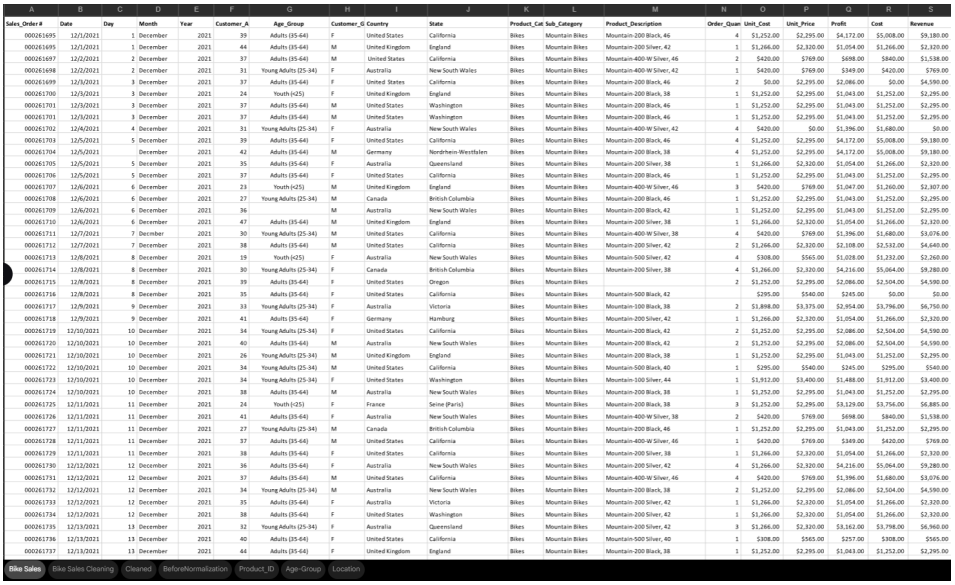
Midterm Lab Task 1. Data Cleaning and Preparation

1. Creating the ERD model

   a. 

2. Unclean Data

   a. 

Sales_Order #	Date	Day	Month	Year	Customer_A	Age_Group	Customer_G	Country	State	Product_Cat	Sub_Category	Product_Description	Order_Quan	Unit_Cost	Unit_Price	Profit	Cost	Revenue
000261695	12/1/2021	1	December	2021	39	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Black, 46	4	$1,252.00	$2,295.00	$4,172.00	$5,008.00	$9,180.00
000261695	12/1/2021	1	December	2021	44	Adults (35-64)	M	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200 Silver, 42	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261697	12/2/2021	2	December	2021	37	Adults (35-64)	M	United States	California	Bikes	Mountain Bikes	Mountain-400-W Silver, 46	2	$420.00	$769.00	$698.00	$840.00	$1,538.00
000261698	12/2/2021	2	December	2021	31	Young Adults (25-34)	F	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W Silver, 42	1	$420.00	$769.00	$349.00	$420.00	$769.00
000261699	12/3/2021	3	December	2021	37	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Black, 46	2	$0.00	$2,295.00	$2,086.00	$0.00	$4,590.00
000261700	12/3/2021	3	December	2021	24	Youth (<25)	F	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200 Black, 38	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261701	12/3/2021	3	December	2021	37	Adults (35-64)	M	United States	Washington	Bikes	Mountain Bikes	Mountain-200 Black, 46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261701	12/3/2021	3	December	2021	37	Adults (35-64)	M	United States	Washington	Bikes	Mountain Bikes	Mountain-200 Black, 46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261702	12/4/2021	4	December	2021	31	Young Adults (25-34)	F	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W Silver, 42	4	$420.00	$0.00	$1,396.00	$1,680.00	$0.00
000261703	12/5/2021	5	December	2021	39	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Black, 46	4	$1,252.00	$2,295.00	$4,172.00	$5,008.00	$9,180.00
000261704	12/5/2021		December	2021	42	Adults (35-64)	M	Germany	Nordrhein-Westfalen	Bikes	Mountain Bikes	Mountain-200 Black, 38	4	$1,252.00	$2,295.00	$4,172.00	$5,008.00	$9,180.00
000261705	12/5/2021	5	December	2021	35	Adults (35-64)	F	Australia	Queensland	Bikes	Mountain Bikes	Mountain-200 Silver, 38	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261706	12/5/2021	5	December	2021	37	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Black, 46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261707	12/6/2021	6	December	2021	23	Youth (<25)	M	United Kingdom	England	Bikes	Mountain Bikes	Mountain-400-W Silver, 46	3	$420.00	$769.00	$1,047.00	$1,260.00	$2,307.00
000261708	12/6/2021	6	December	2021	27	Young Adults (25-34)	M	Canada	British Columbia	Bikes	Mountain Bikes	Mountain-200 Black, 46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261709	12/6/2021	6	December	2021	36		M	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200 Black, 42	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261710	12/6/2021	6	December	2021	47	Adults (35-64)	M	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200 Silver, 38	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261711	12/7/2021	7	December	2021	30	Young Adults (25-34)	M	United States	California	Bikes	Mountain Bikes	Mountain-400-W Silver, 38	4	$420.00	$769.00	$1,396.00	$1,680.00	$3,076.00
000261712	12/7/2021	7	December	2021	38	Adults (35-64)	M	United States	California	Bikes	Mountain Bikes	Mountain-200 Silver, 42	2	$1,266.00	$2,320.00	$2,108.00	$2,532.00	$4,640.00
000261713	12/8/2021	8	December	2021	19	Youth (<25)	F	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-500 Silver, 42	4	$308.00	$565.00	$1,028.00	$1,232.00	$2,260.00
000261714	12/8/2021	8	December	2021	30	Young Adults (25-34)	F	Canada	British Columbia	Bikes	Mountain Bikes	Mountain-200 Silver, 38	4	$1,266.00	$2,320.00	$4,216.00	$5,064.00	$9,280.00
000261715	12/8/2021	8	December	2021	39	Adults (35-64)	F	United States	Oregon	Bikes	Mountain Bikes		2	$1,252.00	$2,295.00	$2,086.00	$2,504.00	$4,590.00
000261716	12/8/2021	8	December	2021	35	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-500 Black, 42		$295.00	$540.00	$245.00	$0.00	$0.00
000261717	12/9/2021	9	December	2021	33	Young Adults (25-34)	F	Australia	Victoria	Bikes	Mountain Bikes	Mountain-100 Black, 38	2	$1,898.00	$3,375.00	$2,954.00	$3,796.00	$6,750.00
000261718	12/9/2021	9	December	2021	41	Adults (35-64)	F	Germany	Hamburg	Bikes	Mountain Bikes	Mountain-200 Silver, 42	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261719	12/10/2021	10	December	2021	34	Young Adults (25-34)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Black, 42	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00	$4,590.00
000261720	12/10/2021	10	December	2021	40	Adults (35-64)	M	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200 Black, 42	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00	$4,590.00
000261721	12/10/2021	10	December	2021	26	Young Adults (25-34)	M	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200 Black, 38	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261722	12/10/2021	10	December	2021	34	Young Adults (25-34)	M	United States	California	Bikes	Mountain Bikes	Mountain-500 Black, 40	1	$295.00	$540.00	$245.00	$295.00	$540.00
000261723	12/10/2021	10	December	2021	34	Young Adults (25-34)	F	United States	Washington	Bikes	Mountain Bikes	Mountain-100 Silver, 44	1	$1,912.00	$3,400.00	$1,488.00	$1,912.00	$3,400.00
000261724	12/10/2021	10	December	2021	38	Adults (35-64)	M	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200 Black, 38	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261725	12/11/2021	11	December	2021	24	Youth (<25)	F	France	Seine (Paris)	Bikes	Mountain Bikes	Mountain-200 Black, 38	3	$1,252.00	$2,295.00	$3,129.00	$3,756.00	$6,885.00
000261726	12/11/2021	11	December	2021	41	Adults (35-64)	F	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W Silver, 38	2	$420.00	$769.00	$698.00	$840.00	$1,538.00
000261727	12/11/2021	11	December	2021	27	Young Adults (25-34)	M	Canada	British Columbia	Bikes	Mountain Bikes	Mountain-200 Black, 46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261728	12/11/2021	11	December	2021	37	Adults (35-64)	M	United States	California	Bikes	Mountain Bikes	Mountain-400-W Silver, 46	1	$420.00	$769.00	$349.00	$420.00	$769.00
000261729	12/11/2021	11	December	2021	38	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Silver, 38	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261730	12/12/2021	12	December	2021	36	Adults (35-64)	F	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200 Silver, 42	4	$1,266.00	$2,320.00	$4,216.00	$5,064.00	$9,280.00
000261731	12/12/2021	12	December	2021	37	Adults (35-64)	M	United States	California	Bikes	Mountain Bikes	Mountain-400-W Silver, 46	4	$420.00	$769.00	$1,396.00	$1,680.00	$3,076.00
000261732	12/12/2021	12	December	2021	34	Young Adults (25-34)	M	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200 Black, 38	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00	$4,590.00
000261733	12/12/2021	12	December	2021	35	Adults (35-64)	F	Australia	Victoria	Bikes	Mountain Bikes	Mountain-200 Silver, 42	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261734	12/12/2021	12	December	2021	38	Adults (35-64)	F	United States	Washington	Bikes	Mountain Bikes	Mountain-200 Silver, 42	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261735	12/13/2021	13	December	2021	32	Young Adults (25-34)	F	Australia	Queensland	Bikes	Mountain Bikes	Mountain-200 Silver, 42	3	$1,266.00	$2,320.00	$3,162.00	$3,798.00	$6,960.00
000261736	12/13/2021	13	December	2021	40	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-500 Silver, 40	1	$308.00	$565.00	$257.00	$308.00	$565.00
000261737	12/13/2021	13	December	2021	44	Adults (35-64)	F	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200 Black, 38	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00

Bike Sales | Bike Sales Cleaning | Cleaned | BeforeNormalization | Product_ID | Age-Group | Location

3. Cleaned Data

a.

Sales_Order #	Date	Day	Month	Year	Customer_Age	Age_Group	Customer_Gender	Country	Lower	Length	Trim	State	Product_Category	Sub_Category	Model	Color	Size
000261695	12/1/2021	1	December	2021	39	Adults (35-64)	Female	United States	united states	13	United States	California	Bikes	Mountain Bikes	Mountain-200	Black	
000261696	12/1/2021	1	December	2021	44	Adults (35-64)	Male	United Kingdom	united kingdom	14	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200	Silver	
000261697	12/2/2021	2	December	2021	37	Adults (35-64)	Male	United States	united states	13	United States	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	
000261698	12/2/2021	2	December	2021	31	Young Adults (25-34)	Female	Australia	australia	9	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	
000261699	12/3/2021	3	December	2021	37	Adults (35-64)	Female	United States	united states	13	United States	California	Bikes	Mountain Bikes	Mountain-200	Black	
000261700	12/3/2021	3	December	2021	24	Youth (<25)	Female	United Kingdom	united kingdom	14	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200	Black	
000261701	12/3/2021	3	December	2021	37	Adults (35-64)	Male	United States	united states	13	United States	Washington	Bikes	Mountain Bikes	Mountain-200	Black	
000261702	12/4/2021	4	December	2021	31	Young Adults (25-34)	Female	United States	united states	13	United States	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	
000261703	12/5/2021	5	December	2021	39	Adults (35-64)	Female	Australia	australia	9	Australia	California	Bikes	Mountain Bikes	Mountain-200	Black	
000261704	12/5/2021	5	December	2021	42	Adults (35-64)	Male	United States	united states	13	United States	Nordrhein-Westfalen	Bikes	Mountain Bikes	Mountain-200	Black	
000261705	12/5/2021	5	December	2021	35	Adults (35-64)	Female	Germany	germany	7	Germany	Queensland	Bikes	Mountain Bikes	Mountain-200	Silver	
000261706	12/5/2021	5	December	2021	37	Adults (35-64)	Female	Australia	australia	9	Australia	California	Bikes	Mountain Bikes	Mountain-200	Black	
000261707	12/6/2021	6	December	2021	23	Youth (<25)	Male	United States	united states	13	United States	England	Bikes	Mountain Bikes	Mountain-400-W	Silver	
000261708	12/6/2021	6	December	2021	27	Young Adults (25-34)	Male	United Kingdom	united kingdom	14	United Kingdom	British Columbia	Bikes	Mountain Bikes	Mountain-200	Black	
000261709	12/6/2021	6	December	2021	36	Youth (<25)	Male	Canada	canada	6	Canada	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	
000261710	12/6/2021	6	December	2021	47	Adults (35-64)	Male	Australia	australia	9	Australia	England	Bikes	Mountain Bikes	Mountain-200	Silver	
000261711	12/7/2021	7	December	2021	30	Young Adults (25-34)	Male	United Kingdom	united kingdom	14	United Kingdom	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	
000261712	12/7/2021	7	December	2021	38	Adults (35-64)	Male	United States	united states	13	United States	California	Bikes	Mountain Bikes	Mountain-200	Silver	
000261713	12/8/2021	8	December	2021	19	Youth (<25)	Female	United States	united states	13	United States	New South Wales	Bikes	Mountain Bikes	Mountain-500	Silver	
000261714	12/8/2021	8	December	2021	30	Young Adults (25-34)	Female	Australia	australia	9	Australia	British Columbia	Bikes	Mountain Bikes	Mountain-200	Silver	
000261715	12/8/2021	8	December	2021	39	Adults (35-64)	Female	Canada	canada	6	Canada	Oregon	Bikes	Mountain Bikes	Mountain-200	Silver	
000261716	12/8/2021	8	December	2021	35	Adults (35-64)	Female	United States	united states	13	United States	California	Bikes	Mountain Bikes	Mountain-500	Black	
000261717	12/9/2021	9	December	2021	33	Young Adults (25-34)	Female	United States	united states	13	United States	Victoria	Bikes	Mountain Bikes	Mountain-100	Black	
000261718	12/9/2021	9	December	2021	41	Adults (35-64)	Female	Australia	australia	9	Australia	Hamburg	Bikes	Mountain Bikes	Mountain-200	Silver	
000261719	12/10/2021	10	December	2021	34	Young Adults (25-34)	Female	Germany	germany	7	Germany	California	Bikes	Mountain Bikes	Mountain-200	Black	
000261720	12/10/2021	10	December	2021	40	Adults (35-64)	Male	United States	united states	13	United States	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	
000261721	12/10/2021	10	December	2021	26	Young Adults (25-34)	Male	Australia	australia	9	Australia	England	Bikes	Mountain Bikes	Mountain-200	Black	
000261722	12/10/2021	10	December	2021	34	Young Adults (25-34)	Male	United Kingdom	united kingdom	14	United Kingdom	California	Bikes	Mountain Bikes	Mountain-500	Black	
000261723	12/10/2021	10	December	2021	34	Young Adults (25-34)	Female	United States	united states	13	United States	Washington	Bikes	Mountain Bikes	Mountain-100	Silver	
000261724	12/10/2021	10	December	2021	38	Adults (35-64)	Male	United States	united states	13	United States	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	
000261725	12/11/2021	11	December	2021	24	Youth (<25)	Female	Australia	australia	9	Australia	Seine (Paris)	Bikes	Mountain Bikes	Mountain-200	Black	
000261726	12/11/2021	11	December	2021	41	Adults (35-64)	Female	France	france	6	France	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	
000261727	12/11/2021	11	December	2021	27	Young Adults (25-34)	Male	Australia	australia	9	Australia	British Columbia	Bikes	Mountain Bikes	Mountain-200	Black	
000261728	12/11/2021	11	December	2021	37	Adults (35-64)	Male	Canada	canada	6	Canada	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	
000261729	12/11/2021	11	December	2021	38	Adults (35-64)	Female	United States	united states	13	United States	California	Bikes	Mountain Bikes	Mountain-200	Silver	
000261730	12/12/2021	12	December	2021	36	Adults (35-64)	Female	United States	united states	13	United States	New South Wales	Bikes	Mountain Bikes	Mountain-200	Silver	
000261731	12/12/2021	12	December	2021	37	Adults (35-64)	Male	Australia	australia	9	Australia	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	
000261732	12/12/2021	12	December	2021	34	Young Adults (25-34)	Male	United States	united states	13	United States	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	
000261733	12/12/2021	12	December	2021	35	Adults (35-64)	Female	Australia	australia	9	Australia	Victoria	Bikes	Mountain Bikes	Mountain-200	Silver	
000261734	12/12/2021	12	December	2021	38	Adults (35-64)	Female	Australia	australia	9	Australia	Washington	Bikes	Mountain Bikes	Mountain-200	Silver	
000261735	12/13/2021	13	December	2021	32	Young Adults (25-34)	Female	United States	united states	13	United States	Queensland	Bikes	Mountain Bikes	Mountain-200	Silver	
000261736	12/13/2021	13	December	2021	40	Adults (35-64)	Female	Australia	australia	9	Australia	California	Bikes	Mountain Bikes	Mountain-500	Silver	
000261737	12/13/2021	13	December	2021	44	Adults (35-64)	Female	United States	united states	13	United States	England	Bikes	Mountain Bikes	Mountain-200	Black	
000261738	12/13/2021	13	December	2021	43	Adults (35-64)	Male	United Kingdom	united kingdom	14	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200	Black	

Bike Sales | Bike Sales Cleaning | Cleaned | BeforeNormalization | Product_ID | Age-Group | Location

4. Before normalization

a.

Sales_Order #	Date	Customer_Age	Age_Group	Customer_Gender	Country	State	Product_Category	Sub_Category	Model	Color	Size	Order_Quantity
000261695	12/1/2021	39	Adults (35-64)	Female	United States	California	Bikes	Mountain Bikes	Mountain-200	Black	46	
000261696	12/1/2021	44	Adults (35-64)	Male	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200	Silver	42	
000261697	12/2/2021	37	Adults (35-64)	Male	United States	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	
000261698	12/2/2021	31	Young Adults (25-34)	Female	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	42	
000261699	12/3/2021	37	Adults (35-64)	Female	United States	California	Bikes	Mountain Bikes	Mountain-200	Black	46	
000261700	12/3/2021	24	Youth (<25)	Female	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200	Black	38	
000261701	12/3/2021	37	Adults (35-64)	Male	United States	Washington	Bikes	Mountain Bikes	Mountain-200	Black	46	
000261702	12/4/2021	31	Young Adults (25-34)	Female	United States	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	42	
000261703	12/5/2021	39	Adults (35-64)	Female	Australia	California	Bikes	Mountain Bikes	Mountain-200	Black	46	
000261704	12/5/2021	42	Adults (35-64)	Male	United States	Nordrhein-Westfalen	Bikes	Mountain Bikes	Mountain-200	Black	38	
000261705	12/5/2021	35	Adults (35-64)	Female	Germany	Queensland	Bikes	Mountain Bikes	Mountain-200	Silver	38	
000261706	12/5/2021	37	Adults (35-64)	Female	Australia	California	Bikes	Mountain Bikes	Mountain-200	Black	46	
000261707	12/6/2021	23	Youth (<25)	Male	United States	England	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	
000261708	12/6/2021	27	Young Adults (25-34)	Male	United Kingdom	British Columbia	Bikes	Mountain Bikes	Mountain-200	Black	46	
000261709	12/6/2021	36	Youth (<25)	Male	Canada	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	42	
000261710	12/6/2021	47	Adults (35-64)	Male	Australia	England	Bikes	Mountain Bikes	Mountain-200	Silver	38	
000261711	12/7/2021	30	Young Adults (25-34)	Male	United Kingdom	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	38	
000261712	12/7/2021	38	Adults (35-64)	Male	United States	California	Bikes	Mountain Bikes	Mountain-200	Silver	42	
000261713	12/8/2021	19	Youth (<25)	Female	United States	New South Wales	Bikes	Mountain Bikes	Mountain-500	Silver	42	
000261714	12/8/2021	30	Young Adults (25-34)	Female	Australia	British Columbia	Bikes	Mountain Bikes	Mountain-200	Silver	38	
000261715	12/8/2021	39	Adults (35-64)	Female	Canada	Oregon	Bikes	Mountain Bikes	Mountain-200	Silver	42	
000261716	12/8/2021	35	Adults (35-64)	Female	United States	California	Bikes	Mountain Bikes	Mountain-500	Black	42	
000261717	12/9/2021	33	Young Adults (25-34)	Female	United States	Victoria	Bikes	Mountain Bikes	Mountain-100	Black	38	
000261718	12/9/2021	41	Adults (35-64)	Female	Australia	Hamburg	Bikes	Mountain Bikes	Mountain-200	Silver	42	
000261719	12/10/2021	34	Young Adults (25-34)	Female	Germany	California	Bikes	Mountain Bikes	Mountain-200	Black	42	
000261720	12/10/2021	40	Adults (35-64)	Male	United States	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	42	
000261721	12/10/2021	26	Young Adults (25-34)	Male	Australia	England	Bikes	Mountain Bikes	Mountain-200	Black	38	
000261722	12/10/2021	34	Young Adults (25-34)	Male	United Kingdom	California	Bikes	Mountain Bikes	Mountain-500	Black	40	
000261723	12/10/2021	34	Young Adults (25-34)	Female	United States	Washington	Bikes	Mountain Bikes	Mountain-100	Silver	44	
000261724	12/10/2021	38	Adults (35-64)	Male	United States	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	38	
000261725	12/11/2021	24	Youth (<25)	Female	Australia	Seine (Paris)	Bikes	Mountain Bikes	Mountain-200	Black	38	
000261726	12/11/2021	41	Adults (35-64)	Female	France	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	38	
000261727	12/11/2021	27	Young Adults (25-34)	Male	Australia	British Columbia	Bikes	Mountain Bikes	Mountain-200	Black	46	
000261728	12/11/2021	37	Adults (35-64)	Male	Canada	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	
000261729	12/11/2021	38	Adults (35-64)	Female	United States	California	Bikes	Mountain Bikes	Mountain-200	Silver	38	
000261730	12/12/2021	36	Adults (35-64)	Female	United States	New South Wales	Bikes	Mountain Bikes	Mountain-200	Silver	42	
000261731	12/12/2021	37	Adults (35-64)	Male	Australia	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	
000261732	12/12/2021	34	Young Adults (25-34)	Male	United States	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	38	
000261733	12/12/2021	35	Adults (35-64)	Female	Australia	Victoria	Bikes	Mountain Bikes	Mountain-200	Silver	42	
000261734	12/12/2021	38	Adults (35-64)	Female	Australia	Washington	Bikes	Mountain Bikes	Mountain-200	Silver	42	
000261735	12/13/2021	32	Young Adults (25-34)	Female	United States	Queensland	Bikes	Mountain Bikes	Mountain-200	Silver	42	
000261736	12/13/2021	40	Adults (35-64)	Female	Australia	California	Bikes	Mountain Bikes	Mountain-500	Silver	40	
000261737	12/13/2021	44	Adults (35-64)	Female	United States	England	Bikes	Mountain Bikes	Mountain-200	Black	38	
000261738	12/13/2021	43	Adults (35-64)	Male	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200	Black	38	

Bike Sales | Bike Sales Cleaning | Cleaned | BeforeNormalization | Product_ID | Age-Group | Location

5. Creating Dim Table

a.

Product_ID	Product_Category	Sub_Category	Model	Color	Size
P-101	Bikes	Mountain Bikes	Mountain-200	Black	46
P-102	Bikes	Mountain Bikes	Mountain-200	Silver	42
P-103	Bikes	Mountain Bikes	Mountain-400-W	Silver	46
P-104	Bikes	Mountain Bikes	Mountain-400-W	Silver	42
P-105	Bikes	Mountain Bikes	Mountain-200	Black	38
P-106	Bikes	Mountain Bikes	Mountain-200	Silver	38
P-107	Bikes	Mountain Bikes	Mountain-200	Black	42
P-108	Bikes	Mountain Bikes	Mountain-400-W	Silver	38
P-109	Bikes	Mountain Bikes	Mountain-500	Silver	42
P-110	Bikes	Mountain Bikes	Mountain-500	Black	42
P-111	Bikes	Mountain Bikes	Mountain-100	Black	38
P-112	Bikes	Mountain Bikes	Mountain-500	Black	40
P-113	Bikes	Mountain Bikes	Mountain-100	Silver	44
P-114	Bikes	Mountain Bikes	Mountain-500	Silver	40
P-115	Bikes	Mountain Bikes	Mountain-200	Silver	46
P-116	Bikes	Mountain Bikes	Mountain-500	Black	44
P-117	Bikes	Mountain Bikes	Mountain-100	Black	48
P-118	Bikes	Mountain Bikes	Mountain-500	Black	52

b.

A	B	C	D	E	F
Age_Gruop_ID	**Age_Group**				
A-101	Adults (35-64)				
A-102	Young Adults (25-34)				
A-103	Youth (<25)				

C.

A	B	C	D	E	F
Location_ID	Country	State			
L-101	United States	California			
L-102	United Kingdom	England			
L-103	Australia	New South Wales			
L-104	United States	Washington			
L-105	United States	New South Wales			
L-106	Australia	California			
L-107	United States	Nordrhein-Westfalen			
L-108	Germany	Queensland			
L-109	United States	England			
L-110	United Kingdom	British Columbia			
L-111	Canada	New South Wales			
L-112	Australia	England			
L-113	United Kingdom	California			
L-114	Australia	British Columbia			
L-115	Canada	Oregon			
L-116	United States	Victoria			
L-117	Australia	Hamburg			
L-118	Germany	California			
L-119	Australia	Seine (Paris)			
L-120	France	New South Wales			
121	Canada	California			
122	Australia	Victoria			
L-123	Australia	Washington			
L-124	United States	Queensland			
L-125	United Kingdom	Washington			
L-126	Australia	Seine et Marne			
L-127	France	Washington			
L-128	Australia	Nordrhein-Westfalen			
L-129	Germany	New South Wales			
L-130	Germany	Seine Saint Denis			
L-131	France	England			
L-132	United States	Seine (Paris)			
L-133	United States	Nord			
L-134	France	Victoria			
L-135	France	Oregon			
L-136	United States	Oregon			
L-137	Australia	South Australia			
L-138	Australia	Hessen			
L-139	Germany	Washington			
L-140	Australia	Queensland			
L-141	Canada	Queensland			
L-142	Australia	Somme			
L-143	France	British Columbia			
L-144	Canada	Hessen			

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